

AC2204 Marking Scheme

Introduction to Management Accounting · Winter 2022 · Paper 1

Examination Session and Year	Winter 2022
Module Code	AC2204
Module Title	Introduction to Management Accounting
Paper Number	Paper 1
External Examiner	Professor Martin Quinn
Head of the Department	Professor Mark Mulcahy
Internal Examiners	Dr Margaret Healy
Duration of Paper	1.5 hours
Special Requirements	Show your workings; marks will be awarded for workings. Students are allowed to use calculators that have standard arithmetic, scientific, graphical or algebraic functions provided they are not programmable and cannot store text (other than algebraic notation).

Instructions to Candidates (as per exam paper): Answer three questions in total. Attempt Question One from Section A (compulsory) and TWO questions from Section B.

Own-figure rule: where a candidate makes an error at an early stage of a multi-step numerical question, full method marks are awarded for correctly applying the required technique to their own (incorrect) figure in all subsequent steps. Marks are never cascaded away for a single early error.

Marking philosophy: this is a closed-book, time-pressured examination. Academic citations are never required. Discursive answers are marked on depth of explanation and application to the specific scenario, not generic textbook recall.

Question Overview

Question	Scenario	Topics	Marks
Q1 (compulsory)	Grove Ltd	CVP analysis, breakeven, margin of safety, operating leverage, multi-product WACM expansion decision, CVP assumptions	40
Q2 (Section B option)	Fraser Ltd	Marginal vs absorption costing, arguments for each method	30
Q3 (Section B option)	Diver Ltd	Traditional (absorption) costing vs activity-based costing, cross-subsidization, batch-level activities	30
Q4 (Section B option)	Statement evaluation	Sunk vs opportunity cost, conversion costs, high-low method, direct/indirect costs, cost structure vs sales mix	30

QUESTION 1**Grove Ltd**

Total: 40 marks

Grove Ltd sells a single product, garden benches, and seeks advice on profit expectations for 2023. Candidates are required to (A) calculate and interpret five core CVP metrics, (B) evaluate a proposed second-product expansion using the weighted average contribution margin approach, and (C) discuss two assumptions underlying CVP analysis.

Part	Requirement	Marks
(A)	Operating profit, breakeven (units and revenue), margin of safety, operating leverage – with explanation of usefulness	20
(B)	WACM breakeven for the two-product expansion; expected profit; advice on proceeding	14
(C)	Two CVP assumptions and their reasonableness	6

Question 1(A)**20 marks**

Grove Ltd sells garden benches: Sales price per unit €240; Variable cost per unit €150; Total fixed expenses €5,400. Assume 75 units will be sold in 2023. Calculate each of the following and explain their usefulness for financial planning: (i) Operating profit (loss); (ii) Breakeven level of sales in units; (iii) Breakeven level of revenues (using the contribution margin approach); (iv) Percentage margin of safety; (v) Operating leverage.

Full Model Answer

Contribution margin per unit = €240 – €150 = €90. Contribution margin ratio = €90 ÷ €240 = 37.5%.

(i) Operating profit - €1,350

Operating profit = (Units sold × contribution per unit) – fixed costs = (75 × €90) – €5,400 = €6,750 – €5,400 = **€1,350**. *Usefulness:* this tells management the expected bottom-line profit at a specific, realistic sales volume, forming the basis for budgeting and assessing whether the venture is financially worthwhile at the projected level of activity.

(ii) Breakeven level of sales in units - 60 units

Breakeven units = Fixed costs ÷ Contribution margin per unit = €5,400 ÷ €90 = **60 units**. *Usefulness:* shows management the minimum sales volume required before the business starts generating a profit, useful for setting minimum sales targets and assessing the riskiness of the venture (75 budgeted units is only 15 units above breakeven).

(iii) Breakeven level of revenues (contribution margin approach) - €14,400

Breakeven revenue = Fixed costs ÷ Contribution margin ratio = €5,400 ÷ 37.5% = **€14,400**. *Usefulness:* expresses the breakeven point in euro sales terms rather than units, which is useful for setting revenue targets and for comparison against budgeted sales revenue directly.

(iv) Percentage margin of safety - 20%

Margin of safety % = (Budgeted sales – Breakeven sales) ÷ Budgeted sales = (75 – 60) ÷ 75 = 15 ÷ 75 = **20%**. *Usefulness:* indicates how far sales can fall from the budgeted level before the company starts making a loss, giving management a measure of the cushion or riskiness built into the sales forecast — a relatively low margin of safety here (20%) signals limited room for a sales shortfall.

(v) Operating leverage - 5.0

Operating (degree of) leverage = Total contribution margin ÷ Operating profit = (75 × €90) ÷ €1,350 = €6,750 ÷ €1,350 = **5.0**. *Usefulness:* indicates how sensitive operating profit is to a change in sales volume — a leverage of 5.0 means a 1% increase in sales revenue would increase operating profit by approximately 5%, highlighting that profit is highly sensitive to sales volume changes at this level of activity (consistent with the relatively low margin of safety above).

Mark Allocation

Marks	Criteria
18-20	All five metrics correctly calculated, with a clear and specific explanation of the usefulness of each for financial planning (not generic statements).
13-17	Four or five metrics correctly calculated with reasonable explanations; one weaker or missing explanation.
8-12	Most metrics correctly calculated but explanations are generic or missing; or a systematic error affects several figures.
0-7	Limited understanding of CVP metrics; minimal correct workings.

Question 1(B)**14 marks**

Grove Ltd is considering expanding by launching Picnic tables: selling price €500/unit, variable cost €100/unit. Additional marketing costs of €4,010 per annum and storage rental of €6,000 per annum would be incurred. Grove Ltd expects to sell 1 Picnic table unit for every 3 Garden bench units sold. (i) Using the Weighted Average Contribution Margin (WACM) approach, determine the breakeven point in units and sales revenues under the expansion proposal. (ii) If actual sales are 75 Garden bench units and 30 Picnic table units, determine the expected operating profit (loss). (iii) Advise Grove Ltd on whether or not to proceed, giving one financial and one non-financial reason.

Full Model Answer - (i) WACM breakeven**Working - revised fixed costs and contribution per unit**

Item	Value
Existing fixed costs	€5,400
Additional marketing costs	€4,010
Storage unit rental	€6,000
Total fixed costs under the expansion proposal	€15,410
Garden bench contribution margin per unit (€240 – €150)	€90
Picnic table contribution margin per unit (€500 – €100)	€400

Working - weighted average contribution margin (sales mix 3 benches : 1 picnic table)

Item	Value
Contribution per 'basket' of 4 units (3 × €90 + 1 × €400)	€670
WACM per unit (€670 ÷ 4)	€167.50

Breakeven point - units and revenue

Item	Value
Total breakeven units (€15,410 ÷ €167.50)	92 units
Garden bench breakeven (92 × 3/4)	69 units
Picnic table breakeven (92 × 1/4)	23 units
Breakeven revenue (69 × €240 + 23 × €500)	€28,060

Alternative revenue approach: Breakeven revenue can equally be found via the weighted average contribution margin ratio: basket revenue of €1,220 (3 × €240 + 1 × €500) gives a WACM ratio of €670 ÷ €1,220 = 54.92%; €15,410 ÷ 54.92% = €28,060, the same result.

Full Model Answer - (ii) Expected operating profit at actual sales

The 3:1 sales mix ratio used in part (i) is only an assumption for the WACM breakeven calculation. Here, actual expected unit sales (75 garden benches and 30 picnic tables) are given directly, so profit is calculated straightforwardly from those actual figures — there is no need to apply the WACM or the assumed 3:1 ratio.

Working - expected operating profit under the expansion

Item	€
Garden bench contribution (75 units × €90)	6,750
Picnic table contribution (30 units × €400)	12,000
Total contribution margin	18,750
Less: Total fixed costs under the expansion proposal	(15,410)
Expected operating profit (with expansion)	3,340

Comparison note: This compares to an operating profit of only €1,350 without the expansion (from part (A)(i)) at the same 75-unit garden bench sales level — the expansion increases expected profit by €1,990, even though the actual sales mix (75:30, i.e. approximately 2.5:1) differs somewhat from the 3:1 mix assumed for the breakeven calculation in part (i).

Full Model Answer - (iii) Advice on the expansion proposal

Financial reason

Proceeding with the expansion is expected to increase operating profit from €1,350 to €3,340, an increase of €1,990 (almost 2.5 times the current profit level) at the stated sales volumes. On this basis alone, the expansion appears financially attractive, provided the sales estimates are reasonably reliable.

Non-financial reason

The initial market research indicating strong demand for picnic tables provides some non-financial support for proceeding, as it reduces (though does not eliminate) the uncertainty around the sales estimates used above. Against this, management should also weigh non-financial risks such as the operational complexity of managing a second product line, reliance on a single rented storage facility, and the fact that market research is inherently uncertain and demand may not materialise as projected.

Overall advice: On balance, given the significant projected profit improvement and supportive market research, Grove Ltd should be advised to proceed with the expansion, while continuing to monitor actual demand closely against the assumptions used, given the relatively low margin of safety identified in part (A).

Mark Allocation

Marks	Criteria
13-14	WACM breakeven correctly calculated in units and revenue; expected profit at actual sales correctly calculated (recognising the actual mix differs from the WACM assumption); clear advice with one distinct, well-reasoned financial and one non-financial factor.
9-12	Substantially correct workings for (i) and (ii) with a minor error; reasonable advice given in (iii).
5-8	WACM approach understood but applied with errors; advice generic or one-sided without balancing considerations.
0-4	Limited understanding of the multi-product breakeven approach; minimal correct workings or advice.

Question 1(C)**6 marks**

Identify and discuss two assumptions underlying CVP analysis. In your answer, consider the extent to which they can be held as reasonable approximations of reality.

Full Model Answer**1. Costs can be reliably classified as fixed or variable, and behave in a linear fashion within the relevant range**

CVP analysis assumes selling price, variable cost per unit, and total fixed costs all remain constant across the range of activity being considered, so total cost and total revenue can be represented as straight lines. In reality this is only an approximation: variable costs per unit may fall due to bulk purchase discounts or rise due to overtime premiums at high volumes, and fixed costs can 'step up' once a certain level of activity is exceeded (e.g. needing a second machine or supervisor). The assumption is reasonable within a limited relevant range of activity over a short time horizon, but becomes less reliable for large changes in volume or over longer periods.

2. The sales mix remains constant (for multi-product businesses)

Where more than one product is sold, CVP/breakeven analysis (as applied using the WACM approach in part (B)) assumes products are sold in a constant, predictable ratio to one another. In reality, sales mix can and does shift over time due to changing customer preferences, seasonality, or promotional activity, which changes the weighted average contribution margin and therefore the breakeven point. This is illustrated directly in part (B) above, where the actual expected sales mix (75 garden benches to 30 picnic tables, approximately 2.5:1) differs from the 3:1 mix assumed for the breakeven calculation — showing that this assumption is often only an approximation of actual trading conditions rather than a precise prediction.

Other acceptable assumptions: Candidates may alternatively discuss: all units produced are sold (no build-up or run-down of inventory, avoiding the fixed-overhead timing distortions seen in marginal vs absorption costing); or that the analysis holds only within a defined relevant range of activity outside of which cost/revenue relationships are assumed to change. Any two well-explained assumptions, discussed with reference to their reasonableness, receive full credit.

Mark Allocation

Marks	Criteria
5-6	Two clearly identified assumptions, each explained accurately and with a genuine, specific discussion of how reasonable the assumption is in practice (not just stating the assumption).
3-4	Two assumptions identified with reasonable explanation; limited discussion of their reasonableness.
1-2	One assumption identified and explained, or two assumptions stated with minimal explanation.
0	No relevant assumptions identified.

QUESTION 1 TOTAL — 40 MARKS

QUESTION 2**Fraser Ltd**

Total: 30 marks

Fraser Ltd commenced operations on 1 January 2020, manufacturing and selling a single product from a premises in Cork city. Candidates are required to (A) prepare income statements under both marginal and absorption costing, and (B) discuss two arguments in support of each method.

Part	Requirement	Marks
(A)	Unit product cost, closing stock value and operating profit – marginal and absorption costing	18
(B)	Two arguments in support of each costing method	12

Question 2(A)**18 marks**

Fraser Ltd commenced operations 1 Jan 2020. Sales 3,600 units; Selling price €26.00; Production 4,000 units; Variable manufacturing expenses (total) €49,200; Fixed manufacturing expenses (total) €16,800; Variable selling expenses (total) €14,400; Fixed selling expenses (total) €11,000. (A) Calculate the unit product cost, the value of closing stock and the operating profit using (i) Marginal costing and (ii) Absorption costing. Show the income statement for each method.

Full Model Answer**Preliminary workings**

Closing stock = Production – Sales = 4,000 – 3,600 = 400 units. Variable manufacturing cost per unit = €49,200 ÷ 4,000 = €12.30. Fixed manufacturing overhead per unit (absorption costing only) = €16,800 ÷ 4,000 = €4.20.

(i) Marginal costing**Unit product cost and closing stock - marginal costing**

Item	€
Unit product cost (variable manufacturing cost only)	12.30
Closing stock value (400 units × €12.30)	4,920

Income statement - marginal costing

Item	€
Sales revenue (3,600 units × €26.00)	93,600
Less: Variable cost of goods sold (3,600 units × €12.30)	(44,280)
Less: Variable selling expenses	(14,400)
Contribution margin	34,920
Less: Fixed manufacturing expenses	(16,800)
Less: Fixed selling expenses	(11,000)
Operating profit (marginal costing)	7,120

(ii) Absorption costing**Unit product cost and closing stock - absorption costing**

Item	€
Unit product cost (€12.30 variable + €4.20 fixed)	16.50
Closing stock value (400 units × €16.50)	6,600

Income statement - absorption costing

Item	€
Sales revenue (3,600 units × €26.00)	93,600
Less: Cost of goods sold (3,600 units × €16.50)	(59,400)
Gross profit	34,200
Less: Variable selling expenses	(14,400)
Less: Fixed selling expenses	(11,000)
Operating profit (absorption costing)	8,800

Reconciliation: The €1,680 profit difference (€8,800 – €7,120) equals the fixed manufacturing overhead carried forward in closing stock under absorption costing (400 units × €4.20 = €1,680), consistent with production (4,000 units) exceeding sales (3,600 units) in this, the company's first year of trading.

Mark Allocation

Marks	Criteria
16-18	Both unit product costs, both closing stock values, and both fully-formatted income statements correct, with contribution margin and gross profit subtotals clearly shown.
12-15	Substantially correct with a minor computational error in one figure, or one method fully correct and the other largely correct.
7-11	Correct approach to at least one costing method; some errors or omissions in workings or presentation.
0-6	Limited understanding of the distinction between the two costing methods; minimal correct workings.

Question 2(B)**12 marks**

Management at Fraser Ltd is undecided about choosing between marginal costing and absorption costing. Discuss two arguments in support of the use of each of the methods.

Full Model Answer**Arguments in support of marginal costing**

1. Better suited to short-term decision-making: by treating fixed costs as period costs and reporting contribution margin separately, marginal costing directly supports CVP analysis, breakeven calculations, and relevant-costing decisions such as special order pricing or make-or-buy decisions, without the distortion of arbitrarily allocated fixed overhead. **2. Removes the incentive to over-produce:** because profit under marginal costing depends only on the number of units sold (not units produced), management cannot artificially inflate reported profit simply by increasing production and building up closing stock, as can happen under absorption costing when fixed overhead is deferred into inventory.

Arguments in support of absorption costing

1. Required for external financial reporting: IAS 2 *Inventories* requires inventory to be valued at the full cost of production (including a fair allocation of fixed production overheads) for external financial statements, so absorption costing must be used for statutory reporting purposes regardless of whether marginal costing is also used internally. **2. Reflects the full cost of production for long-term pricing:** fixed manufacturing overheads are a genuine and necessary cost of running the production facility; absorption costing ensures selling prices are set with reference to the full cost of production (not just variable cost), reducing the risk of underpricing products in the long run and failing to recover fixed overhead costs over time.

Mark Allocation

Marks	Criteria
10-12	Two distinct, well-explained arguments given for each method (four arguments in total), correctly characterising the practical strengths of marginal and absorption costing respectively.
7-9	Good arguments given for both methods; one argument less developed or slightly generic.
4-6	At least one reasonable argument given for each method, or strong arguments for only one method.
0-3	Minimal or no relevant arguments given.

QUESTION 2 TOTAL — 30 MARKS

QUESTION 3**Diver Ltd**

Total: 30 marks

Diver Ltd currently uses traditional (absorption) costing, absorbing production overheads of €1,232,000 on the basis of machine hours across three products. Candidates are required to (A) recalculate product costs under both traditional and activity-based costing, and (B) explain two management accounting terms and their impact on unit cost.

Part	Requirement	Marks
(A)	Total costs and cost per unit – traditional costing and activity-based costing	22
(B)	Product-cost cross-subsidization and batch-level activities	8

Question 3(A)**22 marks**

Diver Ltd absorbs production overheads (€1,232,000) on the basis of machine hours, at €28 per machine hour. Direct labour costs €18 per hour. Product O: 10,000 units, 3.0 DLH/unit, 3.0 MH/unit, 30,000 total MH, €20 DM/unit. Product P: 1,000 units, 1.0 DLH/unit, 2.0 MH/unit, 2,000 total MH, €10 DM/unit. Product S: 3,000 units, 1.0 DLH/unit, 4.0 MH/unit, 12,000 total MH, €16 DM/unit. ABC cost pools: Machining 40% (machine hours), Production scheduling 30% (set-ups), Quality control 20% (units), Materials handling 10% (purchase orders). Purchase orders: O=10, P=10, S=5 (total 25). Set-ups: O=50, P=20, S=50 (total 120). (A) Calculate the total costs and cost per unit for each product using (i) Traditional (absorption) costing and (ii) Activity-based costing.

Full Model Answer - (i) Traditional (absorption) costing

Overhead absorption rate = €1,232,000 ÷ 44,000 total machine hours (30,000 + 2,000 + 12,000) = €28 per machine hour, as given.

Traditional (absorption) costing - cost per unit

Item	Product O	Product P	Product S
Direct materials	20.00	10.00	16.00
Direct labour (DLH × €18)	54.00	18.00	18.00
Overhead (MH/unit × €28)	84.00	56.00	112.00
Total cost per unit	158.00	84.00	146.00

Traditional (absorption) costing - total costs

Item	Product O (€)	Product P (€)	Product S (€)
Units produced	10,000	1,000	3,000
Total cost (unit cost × units)	1,580,000	84,000	438,000

Check: Total overhead absorbed = €840,000 (O) + €56,000 (P) + €336,000 (S) = €1,232,000, agreeing with the total production overhead given.

Full Model Answer - (ii) Activity-based costing

Working - cost pools and activity rates

Cost pool	Pool amount (€)	Cost driver	Total activity	Rate
Machining (40%)	492,800	Machine hours	44,000 MH	€11.20 per MH
Production scheduling (30%)	369,600	No. of set-ups	120 set-ups	€3,080.00 per set-up
Quality control (20%)	246,400	No. of units	14,000 units	€17.60 per unit
Materials handling (10%)	123,200	No. of purchase orders	25 orders	€4,928.00 per order
Total	1,232,000			

Working - overhead allocated to each product

Cost pool	Product O (€)	Product P (€)	Product S (€)
Machining (MH × €11.20)	336,000	22,400	134,400
Scheduling (set-ups × €3,080)	154,000	61,600	154,000
Quality control (units × €17.60)	176,000	17,600	52,800
Materials handling (orders × €4,928)	49,280	49,280	24,640
Total overhead	715,280	150,880	365,840

Check: Total overhead allocated = €715,280 + €150,880 + €365,840 = €1,232,000, agreeing with the total production overhead given.

Activity-based costing - total costs and cost per unit

Item	Product O	Product P	Product S
Direct materials (total)	200,000	10,000	48,000
Direct labour (total)	540,000	18,000	54,000
Overhead (from above)	715,280	150,880	365,840
Total cost	1,455,280	178,880	467,840
Units produced	10,000	1,000	3,000
Cost per unit	145.53	178.88	155.95

Comparison: ABC reduces Product O's unit cost slightly (€158.00 → €145.53) while substantially increasing Product P's (€84.00 → €178.88, more than double) and increasing Product S's (€146.00 → €155.95). Product P, despite being the lowest-volume product, consumes a disproportionate share of set-up and materials-handling activity relative to its machine-hour usage, which the traditional volume-based method fails to capture — this is the essence of the product-cost cross-subsidization tested in part (B).

Mark Allocation

Marks	Criteria
18-22	Traditional costing correctly calculated; all four ABC cost pool rates correctly calculated; overhead correctly allocated to each product; ABC total cost and cost per unit correct for all three products.
13-17	Substantially correct with minor computational errors in one or two pool rates or allocations.
7-12	Traditional costing correct; ABC approach understood but rates or allocations contain multiple errors.
0-6	Limited understanding of either costing method; minimal correct workings.

Question 3(B)**8 marks**

Explain each of the following terms used in management accounting and outline their impact on the cost per unit of a product:

(i) Product-cost cross-subsidization. (ii) Batch-level activities.

Full Model Answer**(i) Product-cost cross-subsidization**

Product-cost cross-subsidization occurs when a costing system's method of allocating overhead causes some products to be overcosted and others undercosted relative to the overhead they actually cause to be incurred — in effect, the overcosted products subsidise the undercosted ones. It typically arises under traditional volume-based costing systems, where a single driver (such as machine hours or direct labour hours) is used to allocate all overhead, even though not all overhead is actually driven by production volume. *Impact on cost per unit:* in this question, traditional costing understates Product P's true cost per unit (€84.00 vs an ABC-based €178.88) because Product P consumes far more set-up and materials-handling activity per unit than its low machine-hour usage suggests, while Product O's cost is correspondingly slightly overstated. This can lead to poor pricing and product-mix decisions, such as underpricing (or over-promoting) the undercosted product and overpricing (or under-promoting) the overcosted one.

(ii) Batch-level activities

Batch-level activities are activities, and their associated costs, that are incurred once for each batch or production run, regardless of how many units are in that batch — for example, machine set-ups, purchase ordering, and batch inspection. Because the cost is fixed per batch rather than varying with the number of units in it, batch-level costs do not vary directly with total production volume. *Impact on cost per unit:* products manufactured in many small batches bear a higher batch-level cost per unit than products manufactured in fewer, larger batches, even if both have identical total production volumes. In this question, the production scheduling (set-up) cost pool is a batch-level cost: Product P requires 20 set-ups for only 1,000 units (a high set-up frequency relative to volume), driving up its ABC cost per unit, whereas a traditional machine-hour-based system spreads this cost evenly regardless of batch frequency, masking the true cost impact of Product P's batching pattern.

Mark Allocation

Marks	Criteria
7-8	Both terms accurately and fully explained, each with a specific, correctly reasoned discussion of the impact on cost per unit that references the question's own figures.
5-6	Both terms explained accurately with reasonable discussion of the impact on cost per unit, but without specific reference to the question's figures.
2-4	One term explained well; the other term explained only partially or generically.
0-1	No meaningful explanation of either term.

QUESTION 3 TOTAL — 30 MARKS

QUESTION 4**Statement Evaluation**

Total: 30 marks

Five independent statements covering sunk vs opportunity cost, conversion cost calculations, the high-low method, direct vs indirect costs, and cost structure vs sales mix. For each statement, candidates must state whether they agree or disagree, with supporting calculations and explanation.

Part	Topic	Marks
(A)	Sunk costs vs opportunity costs	6
(B)	Pigeon Ltd – manufacturing overhead from conversion cost percentage	6
(C)	Pamela's Pet Parlour – high-low method cost estimate	6
(D)	Direct costs and cost objectives	6
(E)	Cost structure vs sales mix	6

Question 4(A)**6 marks**

Sunk costs quantify the potential benefit that is given up when one alternative is selected over another.

Full Model Answer**DISAGREE**

The description given in the statement is the definition of an **opportunity cost**, not a sunk cost.

What a sunk cost actually is

A sunk cost is a cost that has already been incurred in the past and cannot be recovered or altered by any current or future decision. Because it does not differ between the alternatives being considered, a sunk cost is irrelevant to decision-making and should be ignored when evaluating future courses of action.

What the statement is actually describing

The potential benefit given up when one alternative is chosen over another is the definition of an **opportunity cost**. Unlike a sunk cost, an opportunity cost is highly relevant to decision-making, even though it does not appear anywhere in the accounting records, because it represents a genuine economic sacrifice of the next best alternative forgone.

Mark Allocation

Marks	Criteria
5-6	Correct disagreement with accurate definitions of both sunk cost and opportunity cost, clearly identifying that the statement has swapped the two terms.
3-4	Correct disagreement with an accurate definition of one of the two terms.
1-2	Correct disagreement but with vague or partially inaccurate definitions.
0	Incorrect verdict or no meaningful explanation.

Question 4(B)**6 marks**

Pigeon Ltd's manufacturing overhead is 40% of its total conversion costs. If direct labour is €63,000 and direct materials are €45,860, then the manufacturing overhead is €42,000.

Full Model Answer**AGREE**

The manufacturing overhead figure of €42,000 given in the statement is correct.

Key point - direct materials is not used in this calculation

Conversion cost is defined as direct labour plus manufacturing overhead (the cost of converting raw materials into a finished product); it deliberately **excludes** direct materials. The €45,860 direct materials figure given in the statement is therefore not required for this calculation at all — it is additional information not relevant to finding manufacturing overhead from the conversion cost percentage.

Working

Item	Value
Let manufacturing overhead (MOH) = 40% of conversion cost	$MOH = 0.40 \times (DL + MOH)$
$MOH = 0.40 \times €63,000 + 0.40 \times MOH$	$MOH - 0.40 MOH = €25,200$
$0.60 \times MOH = €25,200$	$MOH = €25,200 \div 0.60$
Manufacturing overhead	€42,000

Check

Item	€
Direct labour	63,000
Manufacturing overhead	42,000
Conversion cost	105,000
Manufacturing overhead as % of conversion cost ($€42,000 \div €105,000$)	40% - confirms the given figure

Mark Allocation

Marks	Criteria
5-6	Correct agreement; conversion cost correctly defined as excluding direct materials; algebraic working correctly solved to give €42,000, with a check confirming the 40% relationship.
3-4	Correct agreement and correct final figure, but the exclusion of direct materials from conversion cost not explicitly explained.
1-2	Correct verdict stated but workings contain an error or direct materials incorrectly included in the calculation.
0	Incorrect verdict or no meaningful workings.

Question 4(C)**6 marks**

Activity levels and associated costs for Pamela's Pet Parlour, 2022: Low: April, 150 dogs groomed, total cost €5,320. High: August, 800 dogs groomed, total cost €11,040. If it is planned to groom 200 dogs in January 2023, based on the cost formula for 2022, the expected cost is €3,444.

Full Model Answer**DISAGREE**

The correct expected cost at 200 dogs groomed is €5,760, not the €3,444 stated.

Working - high-low method

Item	Value
Variable cost per dog $((€11,040 - €5,320) \div (800 - 150))$	$(€5,720 \div 650) = €8.80$ per dog
Fixed cost (at high point: $€11,040 - (800 \times €8.80)$)	€4,000
Fixed cost (at low point: $€5,320 - (150 \times €8.80)$) – check	€4,000
Cost function	Total cost = €4,000 + $(€8.80 \times \text{dogs groomed})$

Estimated cost at 200 dogs (January 2023)

Item	Value
$€4,000 + (€8.80 \times 200)$	$€4,000 + €1,760$
Correct estimated cost	€5,760

Note: The €3,444 figure in the statement does not correspond to any obvious alternative calculation from this data and appears to simply be an incorrect figure; candidates are not required to identify the source of the error, only to demonstrate the correct cost function and resulting estimate.

Mark Allocation

Marks	Criteria
5-6	Correct disagreement; variable cost per unit and fixed cost correctly calculated and cross-checked at both points; correct cost function applied to give €5,760.
3-4	Correct disagreement with a minor computational error, or the cross-check at the second point omitted.
1-2	High-low method attempted with the correct general approach but a significant computational error.
0	Incorrect verdict (agrees with the statement) or no meaningful workings.

Question 4(D)**6 marks**

If a given cost is direct with respect to one stated cost objective, then it must be direct with respect to all other cost objectives.

Full Model Answer**DISAGREE**

Whether a cost is direct or indirect is always defined in relation to a **specific, stated cost objective** — the same cost can be direct with respect to one cost objective and indirect with respect to another.

Why directness is objective-specific

A direct cost is one that can be specifically and exclusively traced to a particular cost objective in an economically feasible way. A cost objective can be a product, a department, a customer, a project, or the organisation as a whole, and the same underlying cost can relate to more than one potential cost objective simultaneously. Traceability depends entirely on which cost objective is being considered.

Illustration

A factory supervisor's salary is a direct cost with respect to the cost objective 'the factory/department' (it can be specifically traced to that department), but it is an *indirect* cost with respect to the cost objective 'an individual unit of product' manufactured within that department (since the supervisor oversees many different products and units, and their salary cannot be specifically traced to any one unit without some form of allocation). This shows that the same cost is direct for one cost objective and indirect for another, contradicting the statement.

Mark Allocation

Marks	Criteria
5-6	Correct disagreement with a clear explanation that directness depends on the specific cost objective, supported by a relevant, correctly reasoned example.
3-4	Correct disagreement with reasonable explanation but no clear illustrative example, or an example that is only partially correct.
1-2	Correct verdict stated but with minimal or unclear explanation.
0	Incorrect verdict (agrees with the statement) or no meaningful explanation.

Question 4(E)**6 marks**

Cost structure refers to the relative proportions in which an organisation's products are sold.

Full Model Answer**DISAGREE**

The description given in the statement is the definition of **sales mix**, not cost structure.

What cost structure actually is

Cost structure refers to the relative proportion of fixed costs and variable costs within an organisation's total costs. A company with a cost structure weighted heavily towards fixed costs (capital intensive) will have higher operating leverage and a lower margin of safety than a company with a cost structure weighted towards variable costs (labour intensive), for the same total revenue and total costs, as explored in a similar statement in a previous paper.

What the statement is actually describing

The relative proportions in which an organisation's different products are sold is the definition of **sales mix**. Sales mix is used in multi-product CVP analysis (for example, via the weighted average contribution margin approach) to determine a combined breakeven point across several products, as applied to Grove Ltd's garden bench and picnic table proposal in Question 1. Cost structure and sales mix are both important CVP concepts, but they describe entirely different things: one relates to the fixed/variable split of costs, the other to the mix of products sold.

Mark Allocation

Marks	Criteria
5-6	Correct disagreement with an accurate definition of cost structure and a correct identification that the statement's description actually matches sales mix.
3-4	Correct disagreement with a reasonable definition of cost structure; the sales mix point not explicitly identified.
1-2	Correct disagreement but with an inaccurate or vague definition of cost structure.
0	Incorrect verdict (agrees with the statement) or no meaningful explanation.

QUESTION 4 TOTAL — 30 MARKS